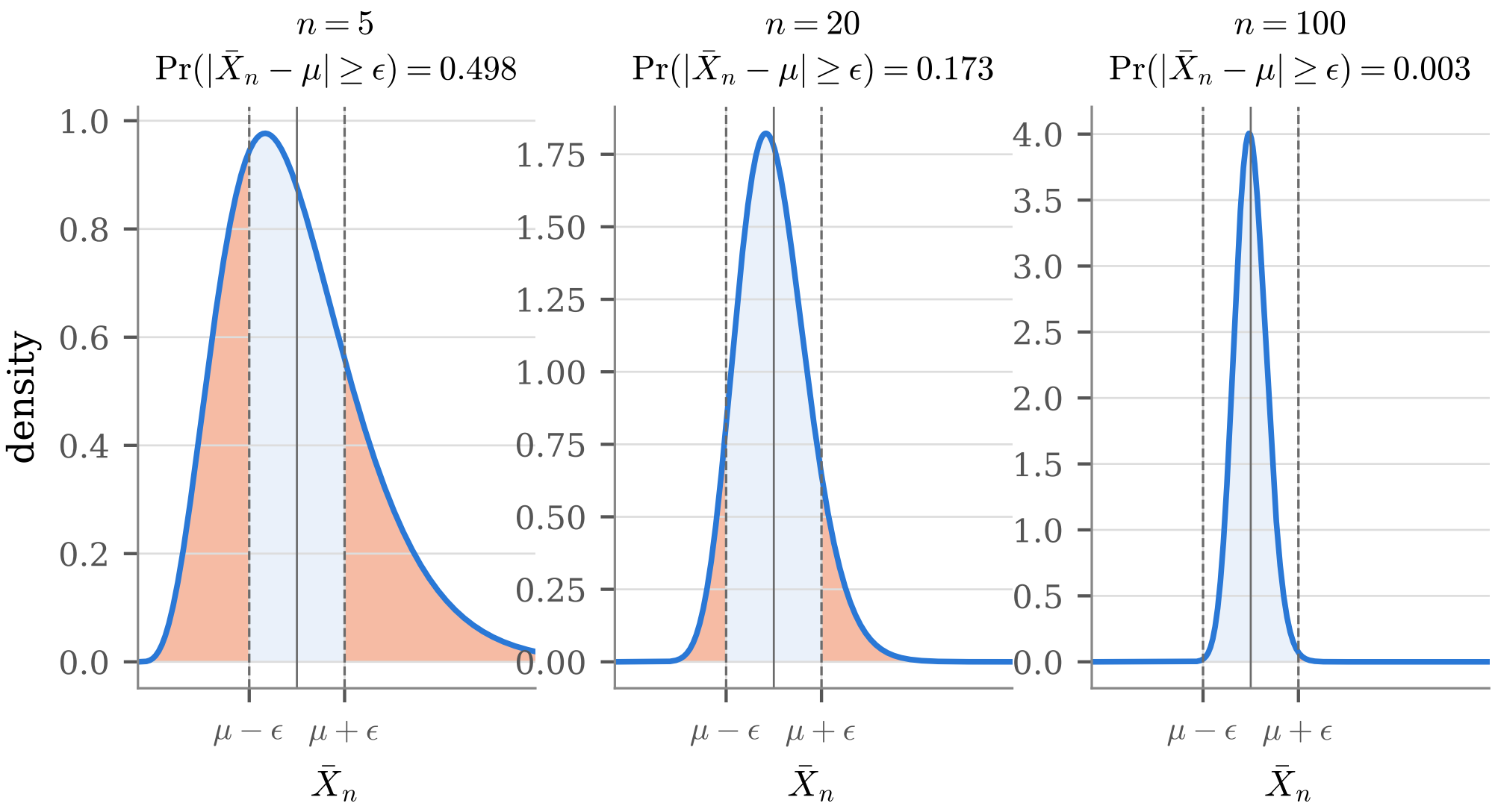

$n = 5$
$\Pr(|\bar{X}_n - \mu| \geq \epsilon) = 0.498$
$n = 20$
$\Pr(|\bar{X}_n - \mu| \geq \epsilon) = 0.173$
$n = 100$
$\Pr(|\bar{X}_n - \mu| \geq \epsilon) = 0.003$
density
1.0
0.8
0.6
0.4
0.2
0.0
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
$\mu - \epsilon$
$\mu + \epsilon$
$\bar{X}_n$